Exception handling

An exception is an event, which occurs during the execution of a program that disrupts the normal flow of the program instructions.

Types of exceptions:

- **Exception(super class for all exceptions in python)**
- **ArthmeticError**
- **OverflowError**
- **ZeroDivisionError**
- **AssertionError**
- **ImportError**
- **IndexError**
- **KeyError**
- **IOError**
- **ValueError**
- **TypeError**
- **NameError**

→if you have some suspicious code that may raise an exception, you can defined you program by placing the suspicious code in a "try:" block.

→after the try block, include "except:" statement, followed by a block of code which handles the problem as elegantly as possible.

Syntax:

```
try:
    program logic
except exception-1:
        executing the code
except exception-2:
      executing the code
……................
……........................
else:
in case of no exception, execute the else block code
```

Example:

```
>>> 12/0
ZeroDivisionError: division by zero
>>> "sushma"+123
TypeError: can only concatenate str (not "int") to str
>>> id=10
>>> print(id1)
NameError: name 'id1' is not defined
```

Example-1:

```
x=input("enter number1:")
y=input("enter number 2:")
try:
   z=int(x)/int(y)
except Exception as e:
   print("exception occured:",e)
print(z)
```

Example-2:

```
import sys
randomList=['a','$',5]
for entry in randomList:
   try:
      print("the entry is",entry)
      r=15/int(entry)
      break
   except:
      ("oops!",sys.exc_info()[0],"occured")
      print("next entry")
```

```
        print()
print("the reciprocal of",entry,"is",r)
```

Example-3:

```
try:
    fh=open("text.txt","r")
    fh.write("this is my test file for exception handling!!!")
except IOError:
    print("error:cannot find file or read data")
else:
    print("written content in the file successfully")
    fh.close()
```

user-define exceptions:

```
class Error(Exception):
    """Base class for other exceptions"""
    pass
class ValueTooSmallError(Error):
    """raised when the input value is too small"""
    pass
class ValueTooLargeError(Error):
    """raised when the input value is too large"""
    pass
number=10
while True:
    try:
        i_num=int(input("enter a number:"))
        if i_num<number:
            raise ValueTooSmallError
        elif i_num>number:
            raise ValueTooLargeError
        break
```

```
    except ValueTooSmallError:
        print("this value is too small, try again!!")
        print()
    except ValueTooLargeError:
        print("this value is too larger,try again!!")
        print()
print("congratulations!!! you guessed it correctly")
```